

STAGE 5 REPORT

Movie Revenue And Audience Trend Analysis

TMDB Movies Dataset | Exploratory Data Analysis

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1. Introduction

The film industry makes billions of dollars every year. But most people do not know that more than half of all movies actually lose money once you account for inflation. Studios spend huge amounts of money deciding which films to make — but most of these decisions are based on gut feeling and past experience rather than looking at what the data actually shows.

This project looks at the TMDB Movies Dataset which contains information on nearly 13 lakh films covering almost 100 years of cinema history. The goal is to use data analysis to find the real patterns behind which films make money and which ones do not. These patterns are shown through three interactive Power BI dashboards — one each for investors, distributors, and producers — so that each group can make better decisions using real data.

1.1 What This Data Revealed That Was Unexpected

Two findings from this dataset were surprising enough to make me check the DAX formulas multiple times before accepting the results as correct.

The first was the **97.85% profit concentration figure**. Going into this analysis, the expectation was that profits in the film industry would be spread somewhat fairly across many films. The data says the complete opposite — just 10% of films earn almost all the money in the entire industry. The remaining 90% of films share only 2.15% of total profit. This number was re-checked three times before it was accepted as real.

The second was the **Mid-Budget Danger Zone**. The natural assumption is that spending more money on a film will give better returns. The data proves this is wrong. Films with budgets between twenty and eighty million dollars give the lowest returns of any budget group. They cost too much to have the creative freedom of a small film but are not big enough to generate the massive audience of a blockbuster. These two surprises shaped every decision made in this analysis.

1.2 The Five Core Discoveries

These five findings come from analysing more than 30,000 films spanning 100 years of cinema history. Every money figure has been adjusted to 2026 dollars using inflation data from the US Bureau of Labor Statistics so that older films and newer films can be compared fairly.

Discovery	Finding	Key Number
One — The Coin-Flip Problem	Only 48.68% of films with complete financial records actually make real profit after adjusting for inflation	48.68%
Two — The Concentration Shock	The top 10% of films control 97.85% of all industry profit. The remaining 90% share only 2.15%	97.85%
Three — The Sleeper Hit Formula	440 films were found that spent under \$10M but earned over \$50M in real profit. Horror films lead this list with 27 confirmed hits	440 films
Four — The Timing Premium	Films released in Summer and December months earn 1.87 times more profit. The gap between December (\$78.54M) and January (\$16.52M) is \$62M just from choosing the right month	1.87×
Five — The Quality Return	Films rated 7.0 or above earn 2.33 times more money. The 7-8 rating group earns \$139.36M average — even more than the 8-10 group	2.33×

1.3 The Central Research Question

"What data-driven strategies help films make the most money?"

Most film studios make decisions about which movies to produce, how much to spend, and when to release them based on experience and instinct rather than data. This project fills that gap directly. By looking at 30,000+ films with complete financial records across nearly 100 years — with all money figures adjusted for inflation — this analysis finds the real patterns behind financial success in cinema. The findings are presented through structured KPIs and interactive Power BI dashboards so that investors, distributors, and producers can all take clear actions based on evidence rather than guesswork.

1.4 Three Stakeholders, Three Questions

Every KPI, every chart, and every finding in this project was built to answer one of three groups' core questions.

Stakeholder	Their Core Question	Dashboard Page
Investors	Is putting money into films a good bet and where is the real risk?	Page 1 — Investor Intelligence

Stakeholder	Their Core Question	Dashboard Page
Distributors	When should a film be released to earn the most money?	Page 2 — Distributor Strategy
Producers	What kind of film should be made, how long, and how much to spend?	Page 3 — Producer Intelligence

2. Dataset Description

Source: TMDB Movies Dataset (2025) on Kaggle

Link: <https://www.kaggle.com/datasets/asaniczka/tmdb-movies-dataset-2023-930k-movies>

Original size: 1,300,000 rows · 24 columns · Coverage: 1900 to 2025

The dataset contains film-level information including production budget, box office revenue, audience ratings, release dates, genres, runtime, and language. The popularity column is a TMDB engagement score. The vote_average column (0 to 10) is the main measure of audience quality used throughout this analysis.

Only films where both budget AND revenue were greater than zero were kept for analysis. These are called **Golden Rows** — 30,000+ films that actually had a cinema release, have real recorded production costs, and can have profit and return on investment calculated properly. These Golden Rows became the base for all analysis from Stage 3 onwards.

All money figures were adjusted to **2026 dollars using CPI data** from the US Bureau of Labor Statistics. Without this step, older films look less profitable than they really were. For example, a \$5M profit in 1980 is worth about \$20.04M in today's money. This adjustment makes sure films from every decade can be compared fairly.

3. Data Preprocessing and Feature Engineering

3.1 Data Cleaning — MIDIO Framework

The MIDIO framework — Missing, Inconsistency, Duplicate, Invalid, Outlier — was applied to every column in the dataset. Each decision was either an ACA (keep or change the data) or a CCA (remove the row or column completely).

Attribute	Issue	Decision	ACA/CCA	Why This Decision Was Made
id	~5,000 duplicates	Removed rows	CCA	Duplicate film IDs create fake entries and break count-based calculations

Attribute	Issue	Decision	ACA/CCA	Why This Decision Was Made
title	18 missing	Removed rows	CCA	A film with no title cannot be identified or analysed
budget	~97.6% were zero	Kept as zero, removed by Golden Row filter	ACA	Cannot estimate 98% of data without creating false results
revenue	~94.9% were zero	Kept as zero, removed by Golden Row filter	ACA	Box office earnings cannot be estimated from other columns
adult	~10,000 inconsistent	Removed rows and column	CCA	Adult content films lack proper box office tracking data
release_date	292,439 missing or wrong format	Converted to proper date format	ACA	Needed to create release_year and release_month columns
genres	595,679 missing	Filled with "Unknown"	ACA	Needed for genre-level analysis for all three stakeholders
tagline	~90.8% missing	Filled with "Unknown"	ACA	Removing these rows would delete 90% of useful financial data
homepage	~61.5% inconsistent URLs	Changed to yes/no flag (has_homepage)	ACA	Web links cannot be used in calculations
poster_path, backdrop_path	All 100% were just image file paths	Removed columns	CCA	Image file paths have no connection to box office earnings
imdb_id, original_title	Repeated information	Removed columns	CCA	TMDB id already works as the unique identifier

No duplicate records were found in the dataset after the id check was completed.

3.2 Feature Engineering

New columns were created from the existing data to allow proper financial analysis. The original dataset had budget and revenue — but those numbers alone do not tell the full story. For example, Film A earned \$200M but cost \$190M giving only \$10M real profit. Film B earned \$50M but cost \$5M giving \$45M profit and a 900% return. Film B was the better investment by every measure.

Financial Columns Created:

New Column	Formula	What It Shows	Who Uses It
profit	revenue – budget	Did the film actually make money?	Investors, Producers
roi_percentage	$(\text{profit} / \text{budget}) \times 100$	Percentage return on every dollar spent	Investors
profit_adj	profit $\times$ CPI number	Real profit in 2026 dollars	All stakeholders
roi_adj	roi $\times$ CPI adjustment	Real return in 2026 dollars	Investors
budget_adj	budget $\times$ CPI number	Real cost in 2026 dollars	Producers
revenue_adj	revenue $\times$ CPI number	Real earnings in 2026 dollars	All stakeholders

Yes/No Columns Created:

New Column	Rule	What It Shows	Who Uses It
is_holiday_release	1 if month is Summer, November or December	Was the film released in holiday season?	Distributors, Producers
is_english	1 if language is English	Was the film made in English?	Distributors
has_homepage	1 if a website link exists	Did the film invest in digital marketing?	Distributors

Time-Based Columns:

Column	Taken From	What It Shows
release_year	release_date	Year the film came out
release_month	release_date	Month the film came out (1 to 12)
decade	release_year	Ten-year grouping for trend analysis

The Sleeper Hit Formula — Original Contribution:

Definition: A Sleeper Hit is any film where **budget_adj** is less than \$10,000,000 AND **profit_adj** is more than \$50,000,000

This formula was created from scratch for this project. Both limits use inflation-adjusted values so a film made in 1975 for \$500K is judged fairly against a 2015 film made for \$8M. The \$50M profit minimum means the film earned at least five times the maximum allowed budget — this is high enough to remove lucky films while catching every genuine success story. This formula is fair across all decades in the dataset.

3.3 Methodology — How The Project Was Done

The full process followed five clear stages:

Raw TMDB Data → Python Cleaning → Inflation Adjustment → Feature Engineering → Correlation Analysis → PCA → DAX Measures → Power BI Dashboard

Why average was not always used:

Films like Avatar, Avengers, and Titanic earn billions. Their earnings pull the average far above what a normal film earns. The middle value (median) gives a more honest picture of what a typical film actually makes. The median profit sitting near zero is one of the most important findings in this project — it shows that most films barely break even in real terms even before counting marketing costs.

A developer can recreate this project by:

1. Downloading the cleaned dataset from the Google Drive link
2. Loading the CSV file into Power BI
3. Creating the DAX measures listed in the KPI section
4. Building the three dashboard pages using the same charts and filters
5. Applying the same colours, labels, and layout used in the final dashboard

Dashboard Overview:

Item	Number
Dashboard Pages	3
Total KPIs	14
DAX Measures Written	18

Item	Number
Filters (Slicers)	10
Films in Analysis	30,000+
Dataset Time Period	1920 to 2026

3.4 Scaling and Dimension Reduction (PCA)

Six number columns — budget, runtime, release_year, popularity, vote_count, vote_average — were scaled using StandardScaler from scikit-learn so all columns contribute equally to the analysis. Revenue, profit, and ROI were left out because they are result columns — including them would make the analysis unfair. Yes/no columns were also left out because they do not work well with PCA calculations.

Four components were kept, together covering 84.38% of total information in the data:

Component	Variance	Total So Far	What It Represents
PC1	37.20%	37.20%	The Blockbuster Dimension — films that score high on all six measures
PC2	21.39%	58.59%	The Modern Quality Shift — newer, well-rated films
PC3	13.91%	72.50%	The Viral Moment — films getting lots of attention right now
PC4	11.88%	84.38%	The Long Film Legacy — older, longer films

4. KPI Identification

Key Performance Indicators were defined for each of the three stakeholder groups. All KPIs were built using DAX formulas in Power BI.

KPI Group 1 — Financial Risk and Return (Investors)

- **KPI 1: Film Success Rate** — What percentage of films with complete financial data actually made real profit after inflation. Value: **48.68%**
- **KPI 2: Average ROI** — Average percentage return on production budget across all Golden Row films. Value: **70.93%** (*this number is misleading because a few blockbusters pull it up — the middle value sits near zero*)
- **KPI 3: Median Adjusted Profit** — The middle profit value across all films after adjusting for inflation. Sits near zero — showing most films barely break even

- **KPI 4: Top 10% Profit Share** — How much of total industry profit the top 10% of films control. Value: **97.85%**
- **KPI 5: Genre Risk-Reward Chart** — A bubble chart showing average return against profit spread by genre, with bubble size showing number of films
- **KPI 6: Sleeper Hit Count** — Number of films meeting the original Sleeper Hit formula. Value: **440 confirmed films**

KPI Group 2 — Release Timing (Distributors)

- **KPI 7: Holiday Release Advantage** — How much more profit holiday releases (November-December) make compared to other months. Value: **1.87 times more**
- **KPI 8: English vs Non-English Revenue Share** — Revenue split by language tracked across each decade from 1900s to 2020s. Non-English share in 2020s: **23.28%**
- **KPI 9: Best and Worst Release Months** — All 12 months ranked by average inflation-adjusted profit. December: **\$78.54M**, January: **\$16.52M**
- **KPI 10: Seasonal Performance** — Average profit grouped into four seasons. Summer leads at **\$55.47M average**

KPI Group 3 — Content and Budget Decisions (Producers)

- **KPI 11: Budget Group Returns** — Average inflation-adjusted return across four budget groups. Low budget leads at **283%**, mid-budget is lowest at **144%**
- **KPI 12: Quality and Revenue Link** — Average inflation-adjusted earnings grouped by audience rating. The 7-8 rating group leads at **\$139.36M**
- **KPI 13: Quality vs Profit Chart** — All 30,000+ films shown as dots by rating (left to right) against profit (up and down), split into four clear zones by two lines — one at rating 7.0 and one at zero profit
- **KPI 14: Film Length and Profit** — Average inflation-adjusted profit grouped by how long the film is. Films over 150 minutes lead at **\$111.38M**

5. Stakeholder-Based Analysis

5.1 Investors — Understanding Financial Risk and Return

What this section is trying to answer:

- What percentage of films actually make money after inflation?
- Is industry profit spread fairly or controlled by a few big films?
- Are there low-cost films that give very high returns?

- Which genres give the best returns for the least risk?

Key Insights:

- Of all 30,000+ films with complete financial data, only **48.68% made real profit** after adjusting for inflation and 51.32% lost money. Picking any single film at random gives roughly the same odds as flipping a coin. This single number tells any investor that putting money into one film is not investing — it is gambling.
- The **top 10% of films control 97.85% of all industry profit**. To picture this clearly — imagine 100 people sharing a pizza. One person eats 97.85 slices and the other 99 people share the remaining 2.15 slices between them. That is the film industry's profit situation. The average ROI of 70.93% looks good but is completely pulled up by a few giant films. The middle value sits near zero.
- **440 Sleeper Hits were confirmed** across the full dataset — only 1.89% of all films. Horror leads with 27, followed by Drama (24), Comedy (17), Documentary (11), and Action (8). If an investor puts \$1M each into ten Horror films — a total of \$10M — and even one becomes a Sleeper Hit earning over \$50M, the whole set of investments is covered and profitable even if the other nine all lose money.
- The genre analysis shows **Horror and Comedy give the best returns for the least risk**. They cost less to make, audiences accept lower production quality as part of the style, and even a complete failure does not wipe out the whole investment.

Impact: Investors can use this data to make smarter decisions about where to put money. Spreading investments across multiple low-budget Horror and Comedy films — backed by 100 years of inflation-adjusted data — gives the best chance of making real returns.

5.2 Distributors — Choosing the Right Release Time

What this section is trying to answer:

- Which months earn the most money for films?
- Does releasing in the holiday season make a real difference?
- Is the non-English film market worth investing in?

Key Insights:

- Films released in ~~November and~~ December earn **1.87 times more profit** than films released in other months. This is not a small seasonal change — it is a consistent advantage confirmed across 100 years of data. Moving a film from any regular month to December costs nothing extra but adds tens of millions in expected profit.
- **December (\$78.54M average profit) and June (\$75.61M)** are the two best months. January is the worst at \$16.52M. The gap between December and January is **\$62 million** — from a pure calendar decision that costs nothing.

Rank	Month	Average Profit	What To Do
1st	December	\$78.54M	Aim for this window for every big release
2nd	June	\$75.61M	Good summer option
3rd	July	\$64.10M	Acceptable second choice
Worst	January	\$16.52M	Avoid for any important release

- Non-English films grew from under 1% of total revenue in the early 1900s to **23.28% in the 2020s**. Films like Parasite from Korea, RRR from India, and Spanish-language productions have shown that non-English films can now reach audiences worldwide. Distributors who do not have non-English film capacity are already missing almost one quarter of all global cinema money.

Impact: Moving a strong film from January to December is worth about \$62 million in expected extra profit at zero extra cost. This is the most valuable and easiest change a distributor can make.

5.3 Producers — Deciding What to Make and How to Make It

What this section is trying to answer:

- How much money should be spent to get the best returns?
- Does making a better quality film actually earn more money?
- How long should a film be for the best commercial results?

Key Insights:

- **Films with budgets under \$20M give 283% average return** — the highest of any budget group. Films with budgets between \$20M and \$80M give only **144% return** despite costing much more. This is the most surprising finding in the whole project. Mid-budget films sit in a danger zone — they cost too much to have the creative freedom of small films but are not big enough to pull in the massive audience of a blockbuster.

Budget Group	Budget Range	Average Return
Low	Under \$20M	283%
Mid	\$20M to \$80M	144%
High	\$80M to \$150M	161%
Very High	Over \$150M	193%

- The **7-8 rating group earns \$139.36M on average** — more than even the 8-10 group which earns only \$57.45M. Films rated between 7 and 8 are liked by a wide mainstream audience and spread through word of mouth. Moving a film from a 6.5 to a 7.5 rating is expected to bring in about **\$54 million more revenue** from quality improvement alone.

Rating Group	Average Earnings	What This Means
Poor (0 to 5)	\$19.14M	Worst result — avoid
Below Average (5 to 6)	\$42.36M	Weak performance
Average (6 to 7)	\$85.23M	Okay but not great
Good (7 to 8)	\$139.36M	Best result — aim for this
Excellent (8 to 10)	\$57.45M	Great quality but smaller audience

- **Films over 150 minutes long earn the most at \$111.38M average profit**, while films under 90 minutes earn only \$15.79M on average. Audiences show through their choices that they are happy to spend more time watching a film they are genuinely enjoying.

Impact: A producer who keeps the budget under \$20M, aims for a 7 to 8 audience rating, and plans a December or June release is following the exact three-factor mix that the data shows gives the best results.

6. Key Findings

1. Film Investment Is Much Riskier Than Most People Think

Only 48.68% of films make real money after inflation. The middle profit value across all films sits near zero. Spreading money across many films is the only approach backed by data.

2. Almost All Profit Goes to a Very Small Number of Films

97.85% of real industry profit goes to just the top 10% of films. This has been true across every decade — it is not a recent change. This makes the industry average return a useless planning number.

3. Low-Budget Films in the Right Genres Can Give Amazing Returns

440 confirmed films prove that spending less than \$10M can lead to over \$50M in real profit. Horror leads this list. The formula works because Horror does not need expensive visual effects and audiences accept the style of lower-cost production.

4. Releasing in December Instead of January Is Worth \$62 Million

December and June are the two best months across 100 years of data. This is a free advantage — it only requires planning and scheduling discipline.

5. Better Quality Directly Means More Money

Films rated 7 or above earn 2.33 times more than lower-rated films. The 7-8 group earns more than even the 8-10 group showing that wide appeal beats niche perfection at the box office.

6. Non-English Films Are the Fastest Growing Part of Global Cinema

Growing from under 1% to 23.28% of global revenue in the 2020s with the growth getting faster each year. This is not slowing down.

7. Recommendations

Recommendation	Who It Is For	Effort Needed	Expected Result
Focus on Horror for low-budget investment	Investors	Medium	Best returns from least spending
Spread money across many films not just one	Investors	Low	Removes the risk of one bad bet wiping everything out
Release in December or June	Distributors	Low	Up to \$62M more profit with no extra cost
Build capacity for non-English films	Distributors	Medium	Capture the fastest-growing part of global cinema
Keep budget under \$20M or go full blockbuster	Producers	Low	Avoid the mid-budget danger zone
Aim for a 7 to 8 audience rating	Producers	Medium	2.33 times more revenue than lower-rated films

Recommendation 1 — Focus on Horror for Low-Budget Film Investment

Horror films have natural advantages shown in the data. They rely on mood and tension rather than expensive effects. Audiences expect and accept simpler visuals as part of the style. A good trailer is enough to market them without big promotional spending. A group of five to ten Horror films each costing under \$10M means just one Sleeper Hit covers everything and puts the whole set of investments into profit.

Recommendation 2 — Spread Money Across Many Films Not Just One

With fewer than half of all films making real money and almost all profit going to unpredictable blockbusters, putting everything into one film has no data support. No more than 15% of total investment should go to any single film. At least six to eight different films across different genres is the minimum safe spread.

Recommendation 3 — Release in December or June

The \$62M gap between the best and worst months is a scheduling choice that costs nothing. Every release plan should calculate how much money is being left behind by choosing any month other than December or June.

Recommendation 4 — Start Building Non-English Film Capacity Now

No decade in the data shows this growth stopping or slowing. At least one non-English film should be part of every production cycle. Subtitling and dubbing infrastructure for Korean, Spanish, and Hindi films — the three biggest growth languages in the data — should be built as soon as possible.

Recommendation 5 — Stay Under \$20M or Go Full Blockbuster

Either make a lean, focused film for under \$20M and aim for the Sleeper Hit zone, or go fully large scale over \$150M where franchise-level audiences justify the spend. The middle range between \$20M and \$80M gives the worst returns of any group. Before approving any film with a budget in that range, the expected return should be compared against what the same money would do split across two or three smaller films.

Recommendation 6 — Aim for a 7 to 8 Audience Rating

Test screenings before final editing decisions should be used to measure expected audience rating. The goal should be above 7.0 before setting a release date. If test results show below 6.5, more editing time is worth more than releasing early into the lowest earning group. The expected difference is \$54M in revenue.

8. Ethical Considerations

The Biggest Limitation — Films That Never Got Tracked

The Golden Rows only include films where both budget and revenue were recorded as greater than zero. Films that never got a proper cinema release, student films, and many regional films are completely missing from this analysis. The 48.68% success rate is almost certainly higher than the real number because the films most likely to be missing are exactly the ones most likely to have failed financially.

Inflation Numbers Are Not Perfect

The CPI data used measures price changes across the whole economy. The film industry may have had different cost changes — for example, the cost of visual effects and cinema tickets may have risen at different rates than general prices. The adjustment used here is correct in direction but not perfectly exact for film-specific costs.

These Are Patterns Not Promises

All the findings describe what has happened across historical data — they do not guarantee the same results in future. There may be other reasons behind the patterns. For example, big studios might release their best films in December AND spend the most money on them — meaning the December result could partly be about budget and quality, not just timing. These recommendations are backed by evidence but should not be treated as guaranteed results.

Data Privacy

The TMDB dataset has no personal information of any kind. Every row represents a film — not a person. No privacy issues apply to any part of this analysis.

Limitations

Limitation	What It Means for Reading the Results
Only about 2% of films had complete financial data	The real failure rate across all films is probably worse than 48.68%
Streaming earnings are not included	Netflix, Amazon and Disney+ film earnings are not in the data — this is a growing gap in the 2020s
Marketing costs are not counted	Real returns are lower than shown because marketing often costs as much as production
Many films belong to more than one genre	Using only the main genre may affect the accuracy of genre-level findings
Inflation data is for the whole economy	Film industry costs may have changed at different rates than general prices
TMDB data is added by volunteers	Older and less well-known films may have missing or wrong information

9. Conclusion

This project looked at 13 lakh films, cleaned the data down to 30,000+ films with complete and reliable financial records, adjusted every money figure for inflation across nearly 100 years, created an original Sleeper Hit formula, ran a full analysis of how financial variables relate to each other, and built a 14-KPI Power BI dashboard across three pages — one for each stakeholder group.

The six most important things this project found:

Finding 1 — The film industry is far more unequal than most people know.

97.85% of real profit going to just 10% of films makes the industry average completely useless as a planning number. Spreading money across many films is the only approach that makes sense.

Finding 2 — Small budget films in the right genres are the smartest investments in cinema.

440 Sleeper Hits confirmed across 100 years. Horror leads the list. The formula uses inflation-adjusted values so it is fair across every decade.

Finding 3 — Choosing the right release month is worth \$62 million.

The gap between December and January is \$62M in expected average profit. This costs nothing — it only requires planning ahead.

Finding 4 — Quality improvements have a clear and measurable money value.

Moving from a 6.5 to a 7.5 audience rating is expected to bring in \$54M more in revenue. Films rated 7 or above earn 2.33 times more than lower-rated films.

Finding 5 — The world film market is changing faster than most people in the industry realise.

Non-English films have grown from under 1% to 23.28% of global revenue. The growth is getting faster not slower.

Finding 6 — Data does not replace creativity. It replaces guesswork.

Every finding in this report came from the data. None of it was invented. The patterns behind which films succeed and which fail are real, consistent, and clear. Using this analysis does not mean giving up creative decisions — it means starting every creative decision from the most honest possible base.

The film industry has treated money decisions as an art form for too long. One hundred years of inflation-adjusted data shows these decisions can also be approached as a science.

Tools Used: Python (Pandas, Scikit-learn, NumPy, Matplotlib, Seaborn) | Microsoft Power BI Desktop | US Bureau of Labor Statistics CPI Data